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Medicament delivery device

Abstract

A medicament delivery device comprises a needle configured to move between a retracted position and an inserted position; an insertion mechanism to move the needle from the retracted position to the inserted position when activated; a locking arm which, in a locked position, is engaged with the insertion mechanism to prevent activation of the insertion mechanism and, in an unlocked position, is disengaged from the insertion mechanism; a retraction mechanism to move the needle from the inserted position to the retracted position when activated; and a trigger configured to move between a disengaged position and an engaged position. On moving from the disengaged position to the engaged position, the trigger is configured to move the locking arm from the locked position to the unlocked position. On moving from the engaged position to the disengaged position, the trigger is configured to activate the retraction mechanism.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of U.S. application Ser. No. 16/321,734, filed on Jan. 29, 2019, which is the national stage entry of International Patent Application No. PCT/EP2017/069135, filed on Jul. 28, 2017, and claims priority to Application No. EP 16182306.7, filed on Aug. 2, 2016, the disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention relates to a device for delivery of medicament to a patient.

BACKGROUND

(2) A variety of diseases exist that require regular treatment by injection of a medicament and such injections can be performed by using injection devices. Injection or infusion pumps of the type known as patch pumps for delivering injections of medicament are known in the art. Another type of injection pump that is gaining traction is the bolus injector device. Some bolus injector devices are intended to be used with relatively large volumes of medicament, typically at least 1 ml and maybe a few ml. Injection of such large volumes of medicament can take some minutes or even hours. Such high capacity bolus injector devices can be called large volume devices (LVDs). Generally, such devices are operated by the patients themselves, although they may also be operated by medical personnel.

(3) To use a patch pump or bolus injector device, such as an LVD, it is first supported on a suitable injection site on a patient's skin. Once installed, injection is initiated by the patient or another person (user). Typically, the initiation is effected by the user operating an electrical switch, which causes a controller to operate the device. Operation includes firstly injecting a needle into the user and then causing the injection of medicament into the user's tissue. Biological medicaments are being increasingly developed which comprise higher viscosity injectable liquids and which are to be administered in larger volumes than long-known liquid medicaments. LVDs for administering such biological medicaments may comprise a pre-filled disposable drug delivery device or, alternatively, a disposable drug delivery device into which a patient or medical personnel must insert a drug cartridge prior to use.

SUMMARY

(4) According to an aspect a medicament delivery device is provided, including a needle configured to move between a retracted position and an inserted position; an insertion mechanism to move the needle from the retracted position to the inserted position when activated; a locking arm which, in a locked position, is engaged with the insertion mechanism to prevent activation of the insertion

mechanism and, in an unlocked position, is disengaged from the insertion mechanism; a retraction mechanism to move the needle from the inserted position to the retracted position when activated; and a trigger configured to move between a disengaged position and an engaged position; wherein, on moving from the disengaged position to the engaged position, the trigger is configured to move the locking arm from the locked position to the unlocked position; and wherein, on moving from the engaged position to the disengaged position, the trigger is configured to activate the retraction mechanism.

(5) The locking arm may be slidably mounted and pivotably coupled to an unlocking lever such that a rotation of the unlocking lever causes a sliding movement of the locking arm.

(6) The trigger, on moving from the disengaged position to the engaged position, may engage with the unlocking lever to rotate the unlocking lever, such that the locking arm is moved from the locked position to the unlocked position.

(7) The unlocking lever may be pivotably mounted at a midpoint, and may include a first end pivotably coupled to the locking arm and a second end arranged to engage with the trigger.

(8) The unlocking lever may be formed having an angle at the pivot point, and may be arranged such that movement of the trigger along a first direction causes a rotation of the first end, and a resulting rotation of the second end causes a sliding movement of the locking arm along a second direction different to the first direction.

(9) The insertion mechanism may include a user actuator arranged to be pushed by a user to activate the insertion mechanism.

(10) The locking arm in the locked position may be engaged with the user actuator to prevent movement of the user actuator.

(11) The locking arm may extend along the direction of sliding motion.

(12) In the locking position, an end of the locking arm may be disposed within a receiving notch formed in the user actuator to prevent movement of the user actuator in a direction perpendicular to the locking arm.

(13) In the unlocked position, the locking arm may be withdrawn from the notch.

(14) The insertion mechanism may include biasing means to urge the needle towards the inserted position; and a blocking element arranged, in a blocking position, to prevent the needle from moving from the retracted position to the inserted position and arranged, in a release position, to allow the needle to move from the retracted position to the inserted position.

(15) The user actuator may be arranged to urge the blocking element from the blocking position to the release position when pushed.

(16) The insertion mechanism may include an injection arm pivotably mounted at one end and pivotably coupled to the needle at the other end; and a guide member configured to restrict movement of the needle to a linear movement between the retracted position and the inserted position in response to a pivoting action of the injection arm.

(17) The blocking element in the blocking position may be arranged in abutment with the injection arm to prevent movement of the needle to the inserted position.

(18) The blocking element may be pivotably mounted to rotate between the blocking position and the release position.

(19) The blocking element may be formed having a substantially triangular shape which is pivotably mounted at a first corner.

(20) A second corner of the blocking element may be in abutment with the injection arm when the blocking element is in the blocking position, and the user actuator may be configured to apply a force at a third corner of the blocking element when pushed.

(21) The biasing means may include a torsion spring.

(22) The torsion spring may be a coil spring fixed around the pivot of the injection arm.

(23) A first end of the coil may be fixed in position, and a second end of the coil may extend along the injection arm to push the injection arm towards the inserted position.

- (24) The retraction mechanism may include a return actuator pivotably mounted to rotate between a first position and a second position.
- (25) The return actuator in the first position may be arranged to abut with the injection arm in the inserted position, and the return actuator may be configured to urge the injection arm and needle to the retracted position of the needle when moved to the second position.
- (26) The return actuator may be mounted to rotate around the pivot point of the injection arm.
- (27) The retraction mechanism may include biasing means configured to urge the return actuator towards the second position.
- (28) The trigger in the engaged state may be configured to engage with the return actuator to retain the return actuator in the first position, and the trigger in the disengaged state may be disengaged from the return actuator.
- (29) The biasing means may include a torsion spring.
- (30) The torsion spring may be a coil spring, having a first end of the coil fixed in position and a second end of the coil in abutment with the return actuator to push the return actuator towards the second position.
- (31) The coil of the torsion spring may not be fixed in position between the first end and second end.
- (32) The blocking element, in the blocking position, may be engaged with the return actuator to retain the return actuator in the first position and, in the release position, may be disengaged from the return actuator.
- (33) The trigger may be formed having a retaining slot.
- (34) In the first position, the return actuator may be aligned with the retaining slot, disposed within the retaining slot and prevented from moving to the second position when the trigger is in the engaged position, and may be released by the retaining slot when the trigger moves from the engaged position to the disengaged position.
- (35) The return actuator, in the second position, may be not aligned with the retaining slot and may abut with the trigger to prevent the trigger from moving from the disengaged position to the engaged position.
- (36) The trigger may be pivotably mounted to rotate between the engaged position and the disengaged position and the trigger may be biased towards the disengaged position.
- (37) The first end of the spring of the retraction mechanism may be fixed to the trigger to push the trigger towards the disengaged position.
- (38) The device may include an outer housing.
- (39) In the retracted position, the needle may be disposed within the housing and, in the inserted position, the needle may extend out of the housing.
- (40) The needle in the inserted position may extend through a contact surface of the housing, and the trigger in the disengaged position may extend through the contact surface.
- (41) The trigger in the disengaged position may extend beyond the needle in the inserted position.
- (42) The trigger in the engaged position may be disposed within the housing, and the trigger may be configured to be moved to the engaged position when the contact surface is placed against a user.
- (43) The trigger may be pivotably mounted to a point within the housing, such that the trigger extends parallel to the contact surface in the engaged position, and extends at an angle from the contact surface in the disengaged position.
- (44) A vertical extent of the housing, measured perpendicular to the contact surface, may be smaller than a horizontal extent of the housing.
- (45) The device may include a medicament reservoir and a medicament which is retained in the medicament reservoir.
- (46) These and other aspects of the invention will be apparent from and elucidated with reference to the embodiments described hereinafter.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Exemplary embodiments of the present invention are described with reference to the accompanying drawings, in which:

(2) FIG. 1 is a schematic side view of an injection device according to an embodiment;

(3) FIG. 2 is an exploded view of the FIG. 1 injection device;

(4) FIG. 3A is a schematic side view of the FIG. 1 injection device;

(5) FIG. 3B is a schematic side view of the FIG. 1 injection device;

(6) FIG. 4A is a schematic side view of the FIG. 1 injection device;

(7) FIG. 4B is a schematic side view of the FIG. 1 injection device;

(8) FIG. 5A is a schematic side view of the FIG. 1 injection device;

(9) FIG. 5B is a schematic side view of the FIG. 1 injection device;

(10) FIG. 5C is a schematic side view of the FIG. 1 injection device;

(11) FIG. 6A is a schematic side view of the FIG. 1 injection device;

(12) FIG. 6B is a schematic side view of the FIG. 1 injection device;

(13) FIG. 6C is a schematic side view of the FIG. 1 injection device;

(14) FIG. 7A is a schematic side view of the FIG. 1 injection device;

(15) FIG. 7B is a schematic side view of the FIG. 1 injection device;

(16) FIG. 8 is a schematic side view of the FIG. 1 injection device;

(17) FIG. 9 is a schematic side view of the FIG. 1 injection device;

(18) FIG. 10 is a schematic side view of the FIG. 1 injection device;

(19) FIG. 11 is a schematic side view of the FIG. 1 injection device; and

(20) FIG. 12 is a schematic side view of the FIG. 1 injection device.

DETAILED DESCRIPTION

(21) A medicament delivery device, as described herein, may be configured to inject a medicament into a patient. For example, delivery could be sub-cutaneous, intra-muscular, or intravenous. Such a device could be operated by a patient or by a care-giver, such as a nurse or physician. The device can include a cartridge-based system that requires piercing a sealed ampule before use. The device can include a large volume device (“LVD”) or patch pump, configured to adhere to a patient's skin for a period of time (e.g., about 5, 15, 30, 60, or 120 minutes) to deliver a “large” volume of medicament (typically about 2 ml to about 10 ml).

(22) In combination with a specific medicament, the presently described devices may also be customized in order to operate within required specifications. For example, the device may be customized to inject a medicament within a certain time period (e.g. about 10 minutes to about 60 minutes for an LVD). Other specifications can include a low or minimal level of discomfort, or to certain conditions related to human factors, shelf-life, expiry, biocompatibility, environmental considerations, etc. Such variations can arise due to various factors, such as, for example, a drug ranging in viscosity from about 3 cP to about 50 cP. Consequently, a drug delivery device will often include a hollow needle ranging from about 25 to about 31 Gauge in size. Common sizes are 27 and 29 Gauge.

(23) The medicament delivery devices described herein can also include one or more automated functions. For example, one or more of needle insertion, medicament injection, and needle retraction can be automated. Energy for one or more automation steps can be provided by one or more energy sources. Energy sources can include, for example, mechanical, pneumatic, chemical, or electrical energy. For example, mechanical energy sources can include springs, levers, elastomers, or other mechanical mechanisms to store or release energy. One or more energy sources can be combined into a single device. Devices can further include gears, valves, or other mechanisms to convert energy into movement of one or more components of a device.

(24) The one or more automated functions of a device may each be activated via an activation mechanism. Such an activation mechanism can include one or more of a button, a lever, a needle sleeve, or other activation component. Activation of an automated function may be a one-step or multi-step process. That is, a user may need to activate one or more activation components in order to cause the automated function. For example, in a one-step process, a user may depress a trigger against their body in order to cause injection of a medicament. Other devices may require a multi-step activation of an automated function. For example, a user may be required to depress a button and deploy a needle in order to cause injection.

(25) In addition, activation of one automated function may activate one or more subsequent automated functions, thereby forming an activation sequence. For example, activation of a first automated function may activate at least two of needle insertion, medicament injection, and needle retraction. Some devices may also require a specific sequence of steps to cause the one or more automated functions to occur. Other devices may operate with a sequence of independent steps.

(26) Some medicament delivery devices can also include one or more functions of a safety syringe, pen-injector, or auto-injector. For example, a delivery device could include a mechanical energy source configured to automatically inject a medicament (as typically found in an auto-injector) and a dose setting mechanism (as typically found in a pen-injector).

(27) FIGS. 1 and 2 show a medicament delivery device **10**, which in the exemplary embodiment comprises a bolus injector device (hereafter simply referred to as “device **10**”), according to a first embodiment. The device **10** comprises a housing **11** containing a medicament delivery mechanism. The device **10** can include an LVD.

(28) FIG. 1 is a schematic cross-section view showing the device **10**. FIG. 2 is an exploded view showing components of the medicament delivery system. The medicament delivery mechanism is only shown schematically and a number of the functional components are omitted for the sake of clarity and brevity, but the device **10** includes a needle **12** for injection of the liquid medicament into a patient's body. The liquid medicament may be provided in a reservoir (not shown) within the medicament delivery mechanism, or may be provided externally of the device **10**.

(29) Although not shown in the figures, a medicament delivery mechanism of a device may include one or more of the following components. A controller configured to control operation of the device **10**. A medicament reservoir containing a supply of medicament to be administered to a patient. The medicament reservoir may, for example, include a cartridge or a vial formed of glass. A plunger maybe provided within the cartridge and plunger driver mechanically coupled to the plunger. The plunger driver may be controllable to move the plunger along the medicament cartridge. The force provided by the plunger causes medicament to be expelled through a medicament delivery aperture in the medicament cartridge and along a medicament delivery tube to the needle **12** to be expelled through the bore of the needle **12**. An electrical power source in the form of a battery to power to the controller. The battery may also provide electrical power the plunger driver, if this is an electrically driven device.

(30) The housing **11** is shown schematically as a rectangular case in FIG. 1. The housing **11** may be a box having a rectangular cross section. Alternatively, the housing **11** may be cylindrical, with the cross section shown through the widest point. Further alternatively, the housing **11** may be approximately dome shaped or may have any other suitable shape and profile to contain the components of the medicament delivery mechanism.

(31) The housing **11** includes a support **16**, as shown in FIG. 2. The support **16** is fixed in position with the housing **11**. The support **16** fixes the positions and movements of the components of the medicament delivery mechanism, as will be described below.

(32) The device **10** generally comprises a housing upper side **11a** and a lower side **11b**. The lower side **11b** is flat. The upper side **11a** may be flat and parallel to the lower side **11b**. Alternatively, the upper side **11a** may be curved or shaped e.g. to be easily held. In an alternative embodiment to lower side **11b** may be curved. The lower side **11b** may be shaped to fit against the patient's skin

when the device **10** is in position.

(33) In use, the lower side **11b** of the housing **11** is intended to be a contact surface that is placed against a patient's skin during a medicament administration process. The lower side **11b** forms a flat contact surface. The lower side **11b** may comprise an adhesive layer to removably adhere to a patient's skin. The lower side **11b** may include a cushioning layer, for example, a foam or gel layer. A height of the device **10**, measured between the upper side **11a** and the lower side **11b**, may be short with respect to the size of the lower side **11b**. A length of the lower side **11b** may be greater than the height of the device **10**. A width of the lower side **11b** may be greater than the height of the device **10**. The device **10** is stable when placed in position against a patient's skin.

(34) The contact surface or lower side **11b** of the housing **11** includes an aperture **13** through which the needle **12** can project in use. The needle **12** is arranged to extend in a direction perpendicular to the lower side **11b**. A length of needle **12** defines a first axis of the device **10**. The first axis is perpendicular to the contact surface with a patient's skin when the lower side **11b** is placed against the patient's skin. The height of the device **10** is measured along the first axis.

(35) A second axis of the device **10** is perpendicular to the first axis. The second axis is parallel to the lower side **11b** of the housing **11**. FIG. 2 shows an exploded view of the device **10**, with components of the medicament delivery system separated along the second axis. In FIG. 1, the second axis extends perpendicular to the plane of the cross section. A third axis of the device **10** is perpendicular to the first axis and the second axis. In FIG. 1, the third axis extends horizontally from left to right.

(36) The needle **12** of the device **10** is movable along the first axis. The needle **12** is prevented from moving except to move along its length. The needle **12** can be moved between a retracted position and an inserted position. In the retracted position the needle **12** is disposed within the housing **11** of the device **10**. In the inserted position, the needle **12** projects from the lower side **11b** of the housing **11** through the aperture **13**. The needle **12** is arranged in the inserted position so as to pierce and inject a patient's skin when the device **10** is attached to a patient.

(37) The needle **12** is fixedly attached to a needle support **12a**. The needle **12** is attached at an upper end to the needle support **12a**. The needle **12** extends downwards from the needle support **12a**. The needle support **12a** is a thin rectangular plate. The needle support **12a** is arranged in a plane perpendicular to the second axis. The needle support **12a** may be any other suitable shape.

(38) The needle **12** and needle support **12a** are arranged to fit within a corresponding groove in the housing support **16**. The groove in the housing support **16** extends vertically along the first axis. The needle support **12a** is in abutment with the sides of the groove. The needle **12** is prevented from moving sideways along the third axis when the needle support **12a** is disposed in the groove.

(39) The needle support **12a** comprises a lateral protrusion. The needle support **12a** has a protrusion extending along the second axis. The protrusion of the needle support **12a** extends away from the housing support **16**. The protrusion extends out of the groove of the housing support **16**. The protruding part of the needle support **12a** is arranged to engage with a needle guide **12b**.

(40) The needle guide **12b** is a generally rectangular frame. The needle guide **12b** is arranged in a plane perpendicular to the second axis. The needle guide **12b** comprises a vertical slot extending along first axis. The protruding part of the needle support **12a** extends into the slot of the needle guide **12b**. The needle support **12a** is in abutment with the sides of the slot of the needle guide **12b**. The needle guide **12b** prevents the needle **12** and the needle support **12a** from sideways movement along the third axis.

(41) The needle guide **12b** is fixed in position against the housing support **16**. The needle support **12a** and the needle **12** are arranged between the housing support **16** and the needle guide **12b**. The needle guide **12b** retains the needle support **12a** in the groove of the housing support **16**. The needle guide **12b** prevents the needle **12** and needle support **12a** from movement along the second axis. The needle **12** is therefore free to move along the first axis only

(42) The medicament delivery mechanism of the device **10** comprises a needle insertion

mechanism **20**, a needle retraction mechanism **30** and a trigger arrangement **40**. The needle insertion mechanism **20** is configured to move the needle **12** from the retracted position into the inserted position. The needle retract mechanism **30** is configured to retract the needle **12** from the inserted position into the retracted position. The trigger arrangement **40** allows the user to activate the medicament delivery process. The trigger arrangement **40** may activate the needle insertion mechanism **20**. Elements of the trigger arrangement **40** may be considered to be part of the needle insertion mechanism **20**.

(43) The device **10** further comprises a contact trigger **14**. The contact trigger **14** is an example of a trigger. The contact trigger **14** is configured to control the operation of the insertion mechanism **20** and the retraction mechanism **30** according to whether or not the device **10** is placed on the body of a patient. Alternatively, the device may include a manual trigger. For example, the trigger may be a grip trigger, to ensure the device is held securely throughout the medicament delivery process.

(44) The contact trigger **14** is configured to lock the insertion mechanism **20** until the device **10** is placed on a contact surface. The contact trigger **14** can prevent activation of the insertion mechanism **20** when the device **10** is not placed on the body. The contact trigger **14** is further configured to activate the retraction mechanism **30** when the device **10** is removed from the body. The contact trigger **14** of the device **10** can prevent unintentional activation of the insertion mechanism **20** when the device **10** is not in position on the body of a patient. The contact trigger **14** also avoids unnecessary exposure of the needle **12** when the medicament delivery process has been completed. The contact trigger **14** can improve the safety of the medicament delivery operation.

(45) The needle insertion mechanism **20** comprises an injection arm **21** and an injection spring **22**. The injection arm **21** is an elongate member. The injection arm **21** is configured to pivot around a point at one end of the injection arm **21**. The injection arm **21** is fixed to the housing support **16** at the pivot point. The injection arm **21** comprises a circular hole at the pivot point, configured to fit over a corresponding axle extending from the housing support **16**. The injection arm **21** freely rotates around the axle of the housing support **16**. The injection arm **21** is arranged to rotate in a plane perpendicular to the second axis.

(46) The needle **12** is coupled to the end of the injection arm **21** distal from the pivot. The injection arm **21** comprises an elongate slot extending along the injection arm **21**. The slot of the injection arm **21** may be open at the distal end of the injection arm **21**, as shown in FIG. 1. Alternatively, the slot of the injection arm **21** may be closed at end, as shown in FIG. 2.

(47) The injection arm **21** is configured to engage with the needle support **12a**. The protruding part of the needle support **12a** passes through the slot of the injection arm **21**. The needle guide **12b** is disposed between the injection arm **21** and the needle **12**. The injection arm **21** extends generally along the third axis, such that rotation about the pivot causes the distal end to move up and down along the first axis. As the injection arm **21** is rotated, the sides of the slot exert a force on the needle support **12a**. The injection arm **21** exerts a force on the needle support **12a** in an upwards or downwards direction along the first axis.

(48) Rotation of the injection arm **21** causes the needle support **12a** to move along the groove of the housing support **16**. Rotation of the injection arm **21** is translated into movement of the needle **12** along the first axis. The injection arm **21** is configured to pivot between a retracted position and an inserted position, corresponding to the retracted position and the inserted position of the needle **12**. The retracted position of the injection arm **21** is shown in FIG. 3A and the inserted position is shown in FIG. 3B.

(49) The injection arm **21** comprises a peg **23**. The peg **23** is arranged to engage with the injection spring **22**. The peg **23** is also for engaging with the trigger arrangement **40**. The peg **23** is positioned midway along injection arm **21**. The peg **23** may be positioned at any point between the pivot and the slot of the injection arm **21**. The peg **23** extends from the injection arm **21** along the second axis. The peg **23** extends in a direction away from the housing support **16**.

(50) The injection spring **22** is an example of a needle driver to drive the needle insertion

mechanism **20**. The injection spring **22** is a torsion spring, for example, a wire coil spring with a straight portion extending from each end of the coil. The ends of the injection spring **22** extend with an angle between them. The injection spring **22** is tensioned so as to reduce the angle between the ends of the coil when released. Alternatively, the needle driver may be, for example, a linear coil spring, a hydraulic or pneumatic piston, or an electric motor. A battery may provide an energy source to power the needle driver. The injection spring **22** is configured to move the needle **12** from the retracted position to the inserted position.

(51) The injection spring **22** is arranged to apply a force which moves the injection arm **21** to the inserted position. The coil of the injection spring **22** is disposed around the pivot point of the injection arm **21**. The coil is disposed around the same axle of the housing support **16** as the injection arm **21**. A first end of the injection spring **22** is fixed in position. A further peg protruding from the housing support **16** is arranged in abutment with the first end of the injection spring **22**. The housing support **16** prevents the first end of the injection spring **22** from moving towards a second end of the injection spring **22**.

(52) The second end of the injection spring **22** is arranged in abutment with the peg **23** of the injection arm **21**. The second end of the injection spring **22** is placed on top of the peg **23**. The peg **23** is positioned between the two ends of the injection spring **22**. The peg **23** prevents the second end of the injection spring **22** from moving towards the first end. The injection spring **22** exerts a downwards force on the peg **23**.

(53) The injection spring **22** can cause a rotation of the injection arm **21**, as shown in FIGS. **4A** and **4B**. When the injection arm **21** is in the retracted position, as shown in FIG. **4A**, the injection spring **22** exerts a downwards force on the peg **23** of the injection arm **21** along the first axis. The injection spring **22** can cause the injection arm **21** to rotate to the inserted position, as shown in FIG. **4B**.

(54) The trigger arrangement **40** comprises a blocking element **41** and a button **42**. The blocking element **41** is coupled between the button **42** and the injection arm **22**. The blocking element **41** comprises a thin, generally triangular element. The blocking element **41** is arranged in a plane perpendicular to the second axis. A first corner of the blocking element **41** forms a pivot point of the blocking element **41**. The blocking element **41** is configured to rotate around the first corner. The blocking element **41** is fixed to the housing support **16** at the pivot point. The blocking element **41** comprises a circular hole positioned at the first corner, configured to fit over a corresponding axle extending from the housing support **16**. The blocking element **41** freely rotates around the axle of the housing support **16**. The blocking element **41** is arranged to rotate in a plane perpendicular to the second axis.

(55) A second corner of the blocking element **41** is configured to engage with the injection arm **21**. The blocking element **41** is configured to engage with the peg **23** of the injection arm **21**. The second corner of the blocking element **41** comprise a cut away portion forming a notch. The pivot point of the blocking element **41** is positioned below the peg **23**. The distance between the first corner and the second corner of the blocking element **41** is configured to equal the distance between the pivot point of the blocking element **41** and the peg **23**.

(56) The blocking element **41** is configured to rotate between a blocking position and a release position, as shown in FIGS. **5A**, **5B** and **5C**.

(57) In the blocking position, as shown in FIG. **5A**, the blocking element **41** is arranged to prevent the injection arm **21** from moving to the inserted position. The blocking element **41** is rotated such that the second corner is above the pivot point. The blocking element **41** engages with the peg **23** to prevent movement of the injection arm **21**. The peg **23** is located in the notch at the second corner of the blocking element **41**. The blocking element **41** prevents the needle **12** from moving to the inserted position in the blocking position.

(58) In the release position, as shown in FIG. **5B**, the blocking element **41** is rotated with respect to the blocking position. The peg **23** is not restrained by the second corner of the blocking element **41**

in the release position. The blocking element **41** allows the injection arm **21** to rotate under the influence of the injection spring **22**. The release position of the blocking element **41** allows the injection arm to move to the inserted position, as shown in FIG. 5C. A side of the blocking element **41** between the first corner and the second corner may be recessed to allow the movement of the peg **23**.

(59) A third corner of the blocking element **41** is configured to be level with the second corner in the blocking position. The blocking element **41** comprises an engaging peg disposed at the third corner. The engaging peg of the blocking element **41** protrudes from the triangular element along the second axis. The engaging peg is arranged to engage with the button **42**.

(60) The button **42** is arranged to be pushed by, for example, a patient or care-giver. The button **42** extends out of the housing **11**. The button **42** extends through the upper side **11a** of the housing **11**. The button **42** is formed with an enlarged upper end. The button **42** may be formed with any shape to be pushed by the user. The button **42** may include a lever arrangement.

(61) The button **42** is configured to move freely along the first axis. A lower portion of the button **42** is in abutment with the engaging peg of the blocking element **41**. A downwards force on the button **42** exerts a corresponding downwards force on the blocking element **41**. A force applied to the upper end of the button **42** therefore causes the blocking element **41** to rotate. The blocking element **41** is moved from a blocking position to a release position when the button **42** is pushed.

(62) The trigger arrangement **40** further comprises a button latch comprising, for example, a locking arm **43** and an unlocking lever **44**. The locking arm **43** is an elongate member. The locking arm **43** is arranged horizontally within the housing **11**. The locking arm **43** extends along the third axis of the device **10**. The locking arm **43** is restricted to move along the third axis only. The locking arm **43** is retained in a horizontal groove formed in the housing support **16**. The upper and lower sides of the locking arm **43** are in abutment with the sides of the groove formed in the housing support **16**. The groove restricts the locking arm **43** from movement along the first axis.

(63) The locking arm **43** is arranged to engage with the button **42** at one end. The button **42** is formed having a receiving notch on one side. An end portion of the locking arm **43** is configured to fit within the receiving notch of the button **42**. When the locking arm **43** is in position in the notch of button **42**, the upper surface of the locking arm **43** is in abutment with the upper side of the receiving notch. The locking arm **43** is restricted from movement along the first axis and can restrict the button **42** from movement along the first axis.

(64) The unlocking lever **44** is a V-shaped element. The unlocking lever **44** is arranged to pivot about a midpoint of the unlocking lever **44**. A first end and a second end of the unlocking lever **44** extend away from the pivot point. An acute angle is formed between the two ends of the unlocking lever **44**. Alternatively, the unlocking lever **44** may be formed having a right angle, or an obtuse angle between the two ends. Further alternatively, the two ends of the unlocking lever **44** may be formed to have different lengths. The unlocking lever **44** is fixed to the housing support **16** at the pivot point. The unlocking lever **44** comprises a circular protrusion configured to fit within a corresponding hole formed in the housing support **16**. The unlocking lever **44** freely rotates in a plane perpendicular to the second axis.

(65) The unlocking lever **44** is coupled to the locking arm **43** at the first end. The first end of the unlocking lever **44** comprises a circular peg extending in the direction of the second axis. The circular peg is configured to fit within a corresponding hole formed in the locking arm **43**. The unlocking lever **44** engages with the locking arm **43** at an end furthest from the button **42**.

Alternatively, the peg formed at the first end of the unlocking lever **44** may engage with a notch formed in a lower surface of the locking arm **43**, as shown in FIG. 1. Further alternatively, the first end of the unlocking lever **44** may be engaged directly with the notch in the locking arm **43**.

(66) The point of engagement between the first end of the unlocking lever **44** and the locking arm **43** is generally above the pivot point of the unlocking lever **44**. Rotation of the unlocking lever **44** results in a movement of the first end of the unlocking lever **44** along the third axis. The unlocking

lever **44** engages with the notch formed in the locking arm **43** when rotated to exert a force on the locking arm **43** along the third axis.

(67) A pivoting movement of the unlocking lever **44** causes a sliding movement of the locking arm **43**. The locking arm **43** and unlocking lever **44** are configured to move between a locked position and an unlocked position. In the locked position, the unlocking lever **44** is engaged with the button **42**. The unlocking lever **44** prevents the button **42** from being pushed in the locked position. In the unlocked position, the unlocking lever **44** is disengaged from the button **42**. The unlocking lever **44** is arranged to engage with the contact trigger **14** at the second end.

(68) The contact trigger **14** is formed as a rectangular arm. The contact trigger **14** extends generally along the third axis of the device **10**. The contact trigger **14** is formed to be relatively thin in the direction of the second axis. The contact trigger **14** is configured to pivot about a point at one end of the rectangular arm. The contact trigger **14** is fixed to the housing support **16** at the pivot point. The contact trigger **14** is formed having a circular peg which is arranged to fit within a corresponding circular hole formed in the housing support **16**. The contact trigger **14** is configured to freely rotate in the circular hole in a plane perpendicular to the second axis.

(69) The contact trigger **14** is movable between a disengaged position and an engaged position, as shown in FIGS. **6A**, **6B** and **6C**. In the engaged position the contact trigger **14** is disposed entirely within the housing **11**. The contact trigger **14** extends horizontally from the pivot point, along the third axis. The contact trigger **14** lies parallel to the lower side **11b**. The lower edge of the contact trigger **14** is above the lower side **11b** of the housing **11**.

(70) In the disengaged position, the contact trigger extends from the lower side **11b** of the housing **11**. The contact trigger **14** is rotated with respect to the engaged position. The contact trigger **14** is angled down from pivot point. The contact trigger **14** extends through a corresponding slot formed in the lower side **11b**. The contact trigger **14** is biased towards the disengaged position. The end of the contact trigger **14** which is furthest from the pivot point forms the lowest point of the contact trigger **14** in the disengaged position. The lowest corner of the contact trigger **14** in the disengaged position may be truncated. The lowest corner may be formed as an edge which is parallel to the lower side **11b** in the disengaged position.

(71) An upper side of the contact trigger **14** is in contact with the unlocking lever **44**. FIG. **6A** shows the contact trigger in an initial state, in the disengaged position. The second end of the unlocking lever **44** comprises a protruding part. The protruding part extends from the unlocking lever **44** along the second axis. The protruding part of the locking arm rests on an upper side of the contact trigger **14**. Alternatively, the protruding part passes through a corresponding hole formed in the contact trigger **14**, as shown in FIG. **2**. Further alternatively, the contact trigger **14** may engage directly with a lower surface of the second end of the unlocking lever **44**.

(72) A force applied upwards along the first axis to the contact trigger **14** exerts a corresponding upwards force on the unlocking lever **44**. The second end of the unlocking lever **44**, which is engaged with the contact trigger **14**, is positioned to the side of the pivot point of the unlocking lever **44**. As the second end of the unlocking lever **44** moves upwards along the first axis, the unlocking lever **44** rotates around the pivot point. The rotation moves the first end of the unlocking lever **44** along the third axis, causing a sliding movement of the locking arm **43** as described above.

(73) FIG. **6B** shows the contact trigger **14** in the engaged position. Movement of the contact trigger **14** from the disengaged position to the engaged position causes rotation of the unlocking lever **44** and moves the locking arm horizontally, as described. The button **42** can be moved along the first axis when the contact trigger **14** is in the engaged position. The button **42** can be pressed to rotate the blocking element **41** as described above, as shown in FIG. **6C**.

(74) The needle retraction mechanism **30** comprises a return actuator **31** and a return spring **32**.

(75) The return actuator **31** comprises an engaging part **33** for engaging with the blocking element **41**, a peg **34** for engaging with the return spring **32** and a locking part **35** for engaging with the contact trigger **14**. The elements of the return actuator **31** are formed in a single piece. The return

actuator **31** is arranged to rotate in a plane perpendicular to the second axis. The return actuator **31** is fixed to the housing support **16** at a pivot point. The return actuator **31** is formed having a circular hole which fits over a corresponding axle extending from the housing support **16**. The return actuator **31** is configured to rotate freely around the axle of the housing support **16**. The return actuator **31** is arranged on the same axle as the injection arm **21** and the injection spring **22**. According to the embodiment, the return actuator **31** is mounted between the injection arm **21** and the housing support **16**.

(76) The engaging part **33** protrudes along second axis from the return actuator **31**. The engaging part **33** extends in the direction of the injection arm **21**. The engaging part **33** is positioned below the injection arm **31**. The engaging part **33** comprises a flat surface for engaging with the lower side of the injection arm **21**.

(77) The peg **34** extends along second axis from the return actuator **31**. The peg **34** extends from the opposite side of the return actuator **31** to the engaging part **33**. The return actuator **31** is arranged between the injection arm **21** and the return spring **32**. The peg **34** is arranged to engage with the return spring **32**.

(78) The return spring **32** is an example of a retraction driver to drive the retraction mechanism **30**. The return spring **32** is a torsion spring, for example, a wire coil spring with a straight portion extending from each end of the coil. The ends of the return spring **32** extend with an angle between them. The return spring **32** is tensioned so as to increase the angle between the ends of the coil when released. Alternatively, the retraction driver may be, for example, a linear coil spring, a hydraulic or pneumatic piston, or an electric motor. A single electric motor may be used to drive the insertion mechanism **20** and the retraction mechanism **30**.

(79) The coil of the return spring **32** is fixed in position with respect to the housing support **16**. The coil is arranged around a peg which extends from the housing support **16** along the second axis. A first end of the return spring **32** is fixed in position. The first end is fixedly attached to the contact trigger **14**, as will be described below. Alternatively, the first end of the return spring **32** may be fixed to an element of the housing **11** or support **16**. The first end of the return spring **32** is prevented from moving away from a second end of the return spring **32**. The second end of the return spring **32** is coupled to the peg **34** of the return actuator **31**. The second end of the return spring **32** is formed in a loop which wraps around the peg **34**. The return spring **32** exerts a force pushing against the peg **34**.

(80) The return spring **32** is configured to move the needle **12** from the inserted position to the retracted position, as shown in FIGS. 7A and 7B. The return actuator **31** is configured to move between a first position and a second position, in response to a force from the return spring **32**. In the first position, as shown in FIG. 7A, the return actuator **31** is positioned below the pivot point. The engaging part **33** is in the lowest position along the first axis. When the injection arm **21** is in the inserted position, the lower side of the injection arm **21** is in abutment with the engaging part **33**.

(81) The return spring **32** is arranged to apply a force which moves the return actuator **31** to the second position. The first end of the return spring **32** is positioned to the side of the return actuator **32**, along the third axis. The tension of the return spring **32** pushes the two ends apart. The return spring **32** exerts a force on the peg **34** along the third axis. The return actuator **32** rotates to the second position in response to the force from the return spring **32**. The return spring **32** pushes the return actuator **32** against the injection arm **21**.

(82) The second position of the return actuator **31** is shown in FIG. 7B. Rotation of the return actuator **32** moves the engaging part **33** along the third axis and upwards along the first axis. The engaging part **33** exerts a corresponding force on the injection arm **21** upwards along the first axis. The injection arm **21** rotates in response to the force from the engaging part **33**. The injection arm **21** is rotated from the inserted position to the retracted position by the movement of the return actuator **31**.

(83) The force exerted by the return spring **32** is larger than the force exerted by the injection spring **22**. The injection arm **21** and the needle **12** will rest in the retracted position when both springs act on the injection arm **21**. Alternatively, the injection arm **21** may be disengaged from the injection spring **22** when moved from the retracted position to the inserted position.

(84) The engaging part **33** of the return actuator **31** is further configured to engage with the blocking element **41**. The engaging part **33** extends downwards from the surface which engages the injection arm **21**. When the return actuator **31** is in the first position, the engaging part **33** extends along the first axis. The engaging part **33** provides a vertically extending surface to engage with the blocking element **41**. The engaging part **33** is arranged to engage a blocking part **41a** of the blocking element **41**.

(85) The blocking part **41a** comprises a rectangular protrusion extending along the second axis from the blocking element **41**. The blocking part **41a** is positioned part way between the first corner and the third corner of the blocking element **41**. The interaction between the engaging part **33** and the blocking part **41a** can be seen in FIGS. 5A and 5B.

(86) When the blocking element **41** is in the blocking position, and the return actuator **31** is in the first position, the engaging part **33** is engaged with the blocking part **41a**. FIG. 5A shows the blocking part **41a** in abutment with the engaging part **33**. The engaging part **33** and the blocking part **41a** are in contact along a vertical surface. The blocking part **41a** cannot move along the third axis due to the fixed pivot point of the blocking element **41**. The engaging part **33** is restrained from moving along the third axis in the direction of the blocking part **33a**. The blocking part **33a** is arranged on the side of the engaging part **33** away from the return spring **32**. The engaging part **33** is pushed against the blocking part **41a** by the return spring **32**. The return actuator **31** is prevented from moving to the second position by the blocking part **41a**.

(87) When the blocking element **41** is rotated to the release position, as shown in FIG. 5B, the blocking part **41a** moves down along the first axis. The blocking part **41a** is moved below the extent of the engaging part **33**. The blocking part **41a** is not in contact with the engaging part **33** in the release position. The engaging part **33** can move past the blocking part **41a** and the return actuator **31** can move to the second position.

(88) The locking part **35** of the return actuator **31** is configured to engage with the contact trigger **14**. The locking part **35** is a rectangular protrusion which extends from the return actuator **31** along the second axis. The locking part **35** extends from the other side of the return actuator **31** to the engaging part **33** and the peg **34**. The locking part **35** is elongate along the first axis. The locking part **35** provides a vertically extending surface to engaging with the contact trigger **14**. The locking part **35** is configured to engage with a corresponding notched portion of the contact trigger **14**.

(89) An interaction between the locking part **35** and the contact trigger **14** is shown in FIGS. 8A, 8B and 8C.

(90) An upper surface of the contact trigger **14** is formed to have a rectangular slot **15**. The shape of the contact trigger corresponds to the shape of the locking part **35**. An upper section of the slot **15** may be enlarged to include an additional circular portion. FIG. 8A shows the locking part **35** engaged with the contact trigger **14**. The return actuator **31** is in the first position and the contact trigger **14** is in the engaged position. The locking part **35** is in abutment with the vertical sides of the slot **15**. The locking part **35** is restrained from moving along the third axis by the side wall of the slot **15**. The return spring **32** pushes the locking part **35** against the side of the slot **15**. The return actuator **31** is restrained from moving to the second position.

(91) FIG. 8B shows the contact trigger **14** in the disengaged position. The contact trigger **14** is moved down with respect to the locking part **35**. The locking part **35** is not in contact with the side walls of the slot **15**. The contact trigger **14** is moved such that the locking part **35** is aligned with the circular portion formed in the contact trigger **14**. The locking part **35** can move along the third axis. The return actuator **31** can be rotated to the second position, as shown in FIG. 8C.

(92) The return spring **32** is configured to bias the contact trigger **14** to the disengaged position.

The first end return spring **32** is fixedly attached to the contact trigger **14**. The first end of the return spring **32** forms a loop around a corresponding protrusion on the surface of the contact trigger **14**. The return spring **32** is attached to the contact trigger **14** at a point below the pivot point of the contact trigger **14**. The return spring **32** exerts a force on the contact trigger **14** along the third axis. The return spring **32** rotates the contact trigger **14** from the engaged position to the disengaged position. The contact trigger **14** can be moved to the engaged position on contact with the skin of a patient.

(93) Operation of the contact trigger **14** and the medicament delivery mechanism will be described in detail with respect to FIGS. **8** to **12**.

(94) FIG. **1** shows the device **10** in an initial state. The contact trigger **14** is in the disengaged position. The locking arm **43** is in the locked position. The button **42** is locked. The blocking element **41** is in the blocking position. The injection arm **21** and the needle **12** are in the retracted position. The return actuator **31** is in the first position.

(95) FIG. **8** shows the device **10** in an engaged state. The device **10** is shown in position on the skin of a patient. The device **10** may be placed in position by a patient themselves or by a care-giver. Contact with the skin of a patient causes the contact trigger **14** to move from the disengaged position to the engaged position.

(96) The contact trigger **14** is engaged with the second end of the unlocking lever **44**. Movement of the contact trigger **14** to the engaged position causes the unlocking lever **44** and locking arm **42** to move from the locked position to the unlocked position. The trigger mechanism **40** of the device **10** can be unlocked by movement of the contact trigger **14** to the engaged position. The trigger mechanism **40** can be unlocked by placing the lower side **11b** of the device **10** in position on the skin of a patient. In the engaged state, the button **42** can be pushed by a patient or care-giver to activate the insertion mechanism **20**.

(97) The insertion mechanism **20** of the device **10** cannot be activated when the device **10** is not engaged with the skin of a patient. The device **10** avoids activation of the insertion mechanism **20** and deployment of the needle before the device **10** is properly attached to a patient. Additionally, when the contact trigger **14** is moved to the engaged position the trigger mechanism **40** can be automatically unlocked. The device **10** can provide an operation which is safer and more convenient for the patient or care-giver operating the device.

(98) The engaging part **33** of the return actuator **31** is configured to engage with the slot **15** when the contact trigger is in the engaged position. The slot **15** is configured to prevent the return actuator **31** from moving to the second position. As shown in FIG. **8**, the return actuator **31** is prevented from moving to the second position by the blocking element **41** and by the contact trigger **14**.

(99) FIG. **9** shows the device **10** in a first activated state. The button **42** is pushed down. The button **42** engages with the blocking element **41** when pushed to move the blocking element **41** from the blocking position to the release position. The blocking element **41** is disengaged from the peg **23** of the injection arm **21**. The injection arm **21** is free to move to the inserted position. The blocking element **41** is further disengaged from the return actuator **31**. The return actuator **31** is retained in the first position by the contact trigger **14**.

(100) FIG. **10** shows the device **10** in a second activated state. The needle **12** is moved to the inserted position. The injection arm **21** is moved to the inserted position by the injection spring **22**. The injection arm **21** pivots until the peg **23** engages with the blocking element **41**. The blocking element **41** is configured to limit the movement of the injection arm **21** in the inserted position. The injection arm **21** moves the needle **12** through the aperture **13** and causes the needle **12** to be injected into the skin of a patient. Once the needle insertion mechanism **20** has been activated, the medicament delivery process can begin to delivery medicament through the needle **12**. The medicament delivery process may be activated manually by a patient or care-giver. Alternatively, the medicament delivery process may be activated automatically following the activation of the

needle insertion mechanism **20**.

(101) The engaging part **33** of the return actuator **31** is disposed in the slot **15** of the contact trigger **14**. The return actuator **31** is retained in the first position. When the insertion mechanism **20** has been activated, the retraction mechanism **30** can be activated according to the position of the contact trigger **14**. The contact trigger **14** is configured to activate the retraction mechanism **30** when the device **10** is removed from the body of a patient.

(102) FIG. **11** shows the device **10** is a first completed state. The medicament delivery process is completed and the device **10** is detached from the skin of the patient. The contact trigger **14** is moved from the engaged position to the disengaged position by the return spring **32**. The contact trigger **14** extends from the lower side **11b** of the housing **11** in the disengaged position. As the device **20** is lifted away from the skin of a patient, the contact trigger **14** is maintained in contact with the skin by the return spring **32** until the disengaged position is reached. The contact trigger **14** can cover the needle **12** as the device **10** is detached from a patient. The contact trigger **14** can protect the needle **12** until the retraction mechanism **30** is activated.

(103) In the completed state of the device **10**, the engaging part **33** is disengaged from the slot **15**. The return actuator **31** is free to move to the second position. The return spring **32** is configured to move the return actuator **31** to the second position. The retraction mechanism **30** can be automatically activated by the contact trigger **14** when the device **10** is detached from the skin of a patient.

(104) FIG. **12** shows the device **10** in a second completed state. The return actuator **31** has been moved to the second position by the return spring **32**. The return actuator **31** engages with the injection arm **21** and moves the injection arm **21** to the retracted position. The needle **12** is moved to the retracted position by the retraction mechanism **30**. The engaging part **33** of the return actuator **31** is arranged to engage with the contact trigger **14** when the return actuator is in the second position. The engaging part **33** is configured to prevent the contact trigger **14** from moving to the engaged position. The contact trigger **14** is maintained in the disengaged position by the engaging part **33** when the device **10** has been used. The device **10** provides a clear indication that the medicament delivery process is completed, and avoids an attempt to use the device **10** a second time.

(105) The retraction mechanism **30** of the device **10** can be activated automatically when the contact trigger **14** moves to the disengaged position. The needle **12** can be automatically moved to the retracted position when the medicament delivery process is completed. The device **10** can avoid injury caused by exposure of the needle **12** when the device **10** is detached from a patient. The device **10** can provide an operation which is safer and more convenient for the patient or care-giver operating the device.

(106) Although a few embodiments have been shown and described, it will be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the invention, the scope of which is defined in the appended claims. Various components of different embodiments may be combined where the principles underlying the embodiments are compatible.

(107) For example, the user activated button may be any form of user actuated input, for instance a switch or lever. The button may be an electronic touch sensor or a soft button provided on a touchscreen. The button may be a component of the needle insertion mechanism. The button may be arranged to push the needle directly from the retracted position to the inserted position. The blocking element may be slidably mounted and configured to slide from the blocking position to the release position in response to, for example, a pushing force from the button or a rotating action of a user actuated lever.

(108) The locking arm may be arranged to engage with the blocking element to prevent movement of the blocking element from the blocking position to the release position. Alternatively, the locking arm may engage with the injection arm or the needle support directly, such that the button can be pushed with no effect until the needle insertion mechanism is unlocked. The button may

include a biasing means to reset the position when pushed and released. Further alternatively, the locking arm may replace the blocking element such that the injection arm is released when the locking arm is disengaged. In this way, the needle insertion mechanism may be activated by the contact trigger.

(109) The contact trigger may be a linear arrangement configured to move perpendicularly out of the lower side of the housing. A coil spring may be arranged to urge the contact trigger along a linear axis. Alternatively, the contact trigger may be a button or lever mounted on the lower side of the case, or may be provided by a proximity or contact sensor.

(110) The needle insertion mechanism may be provided by a coil spring or another linear actuator, rather than the pivoting arrangement described above. A spring or linear needle driver may be mounted on the axis of the needle to drive the needle directly into the inserted position. The injection spring and return spring may be formed by any suitable biasing means, for example, a compression coil spring, a tension coil spring, a flat spring or by an elastic (e.g. rubber or polymer) element. Alternatively, the needle insertion mechanism and needle retraction mechanism may be driven by any suitable linear or rotary actuator such as a motor, an electromagnetic driver or a pneumatic or hydraulic driver.

(111) A plurality of needles may be provided in an array for injection by the needle insertion mechanism, or a plurality of needle insertion mechanisms may be provided.

(112) The terms “drug” or “medicament” are used synonymously herein and describe a pharmaceutical formulation containing one or more active pharmaceutical ingredients or pharmaceutically acceptable salts or solvates thereof, and optionally a pharmaceutically acceptable carrier. An active pharmaceutical ingredient (“API”), in the broadest terms, is a chemical structure that has a biological effect on humans or animals. In pharmacology, a drug or medicament is used in the treatment, cure, prevention, or diagnosis of disease or used to otherwise enhance physical or mental well-being. A drug or medicament may be used for a limited duration, or on a regular basis for chronic disorders.

(113) As described below, a drug or medicament can include at least one API, or combinations thereof, in various types of formulations, for the treatment of one or more diseases. Examples of API may include small molecules having a molecular weight of 500 Da or less; polypeptides, peptides and proteins (e.g., hormones, growth factors, antibodies, antibody fragments, and enzymes); carbohydrates and polysaccharides; and nucleic acids, double or single stranded DNA (including naked and cDNA), RNA, antisense nucleic acids such as antisense DNA and RNA, small interfering RNA (siRNA), ribozymes, genes, and oligonucleotides. Nucleic acids may be incorporated into molecular delivery systems such as vectors, plasmids, or liposomes. Mixtures of one or more drugs are also contemplated.

(114) The term “drug delivery device” shall encompass any type of device or system configured to dispense a drug or medicament into a human or animal body. Without limitation, a drug delivery device may be an injection device (e.g., syringe, pen injector, auto injector, large-volume device, pump, perfusion system, or other device configured for intraocular, subcutaneous, intramuscular, or intravascular delivery), skin patch (e.g., osmotic, chemical, micro-needle), inhaler (e.g., nasal or pulmonary), an implantable device (e.g., drug- or API-coated stent, capsule), or a feeding system for the gastro-intestinal tract. The presently described drugs may be particularly useful with injection devices that include a needle, e.g., a hypodermic needle for example having a Gauge number of 24 or higher.

(115) The drug or medicament may be contained in a primary package or “drug container” adapted for use with a drug delivery device. The drug container may be, e.g., a cartridge, syringe, reservoir, or other solid or flexible vessel configured to provide a suitable chamber for storage (e.g., short- or long-term storage) of one or more drugs. For example, in some instances, the chamber may be designed to store a drug for at least one day (e.g., 1 to at least 30 days). In some instances, the chamber may be designed to store a drug for about 1 month to about 2 years. Storage may occur at

room temperature (e.g., about 20° C.), or refrigerated temperatures (e.g., from about -4° C. to about 4° C.). In some instances, the drug container may be or may include a dual-chamber cartridge configured to store two or more components of the pharmaceutical formulation to-be-administered (e.g., an API and a diluent, or two different drugs) separately, one in each chamber. In such instances, the two chambers of the dual-chamber cartridge may be configured to allow mixing between the two or more components prior to and/or during dispensing into the human or animal body. For example, the two chambers may be configured such that they are in fluid communication with each other (e.g., by way of a conduit between the two chambers) and allow mixing of the two components when desired by a user prior to dispensing. Alternatively, or in addition, the two chambers may be configured to allow mixing as the components are being dispensed into the human or animal body.

(116) The drugs or medicaments contained in the drug delivery devices as described herein can be used for the treatment and/or prophylaxis of many different types of medical disorders. Examples of disorders include, e.g., diabetes mellitus or complications associated with diabetes mellitus such as diabetic retinopathy, thromboembolism disorders such as deep vein or pulmonary thromboembolism. Further examples of disorders are acute coronary syndrome (ACS), angina, myocardial infarction, cancer, macular degeneration, inflammation, hay fever, atherosclerosis and/or rheumatoid arthritis. Examples of APIs and drugs are those as described in handbooks such as Rote Liste 2014, for example, without limitation, main groups 12 (anti-diabetic drugs) or 86 (oncology drugs), and Merck Index, 15th edition.

(117) Examples of APIs for the treatment and/or prophylaxis of type 1 or type 2 diabetes mellitus or complications associated with type 1 or type 2 diabetes mellitus include an insulin, e.g., human insulin, or a human insulin analogue or derivative, a glucagon-like peptide (GLP-1), GLP-1 analogues or GLP-1 receptor agonists, or an analogue or derivative thereof, a dipeptidyl peptidase-4 (DPP4) inhibitor, or a pharmaceutically acceptable salt or solvate thereof, or any mixture thereof. As used herein, the terms “analogue” and “derivative” refer to any substance which is sufficiently structurally similar to the original substance so as to have substantially similar functionality or activity (e.g., therapeutic effectiveness). In particular, the term “analogue” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, by deleting and/or exchanging at least one amino acid residue occurring in the naturally occurring peptide and/or by adding at least one amino acid residue. The added and/or exchanged amino acid residue can either be codeable amino acid residues or other naturally occurring residues or purely synthetic amino acid residues. Insulin analogues are also referred to as “insulin receptor ligands”. In particular, the term “derivative” refers to a polypeptide which has a molecular structure which formally can be derived from the structure of a naturally occurring peptide, for example that of human insulin, in which one or more organic substituent (e.g. a fatty acid) is bound to one or more of the amino acids. Optionally, one or more amino acids occurring in the naturally occurring peptide may have been deleted and/or replaced by other amino acids, including non-codeable amino acids, or amino acids, including non-codeable, have been added to the naturally occurring peptide.

(118) Examples of insulin analogues are Gly(A21), Arg(B31), Arg(B32) human insulin (insulin glargine); Lys(B3), Glu(B29) human insulin (insulin glulisine); Lys(B28), Pro(B29) human insulin (insulin lispro); Asp(B28) human insulin (insulin aspart); human insulin, wherein proline in position B28 is replaced by Asp, Lys, Leu, Val or Ala and wherein in position B29 Lys may be replaced by Pro; Ala(B26) human insulin; Des(B28-B30) human insulin; Des(B27) human insulin and Des(B30) human insulin.

(119) Examples of insulin derivatives are, for example, B29-N-myristoyl-des(B30) human insulin, Lys(B29) (N-tetradecanoyl)-des(B30) human insulin (insulin detemir, Levemir®); B29-N-palmitoyl-des(B30) human insulin; B29-N-myristoyl human insulin; B29-N-palmitoyl human insulin; B28-N-myristoyl LysB28ProB29 human insulin; B28-N-palmitoyl-LysB28ProB29 human

insulin; B30-N-myristoyl-ThrB29LysB30 human insulin; B30-N-palmitoyl-ThrB29LysB30 human insulin; B29-N—(N-palmitoyl-gamma-glutamyl)-des(B30) human insulin, B29-N-omega-carboxypentadecanoyl-gamma-L-glutamyl-des(B30) human insulin (insulin degludec, Tresiba®); B29-N—(N-lithocholyl-gamma-glutamyl)-des(B30) human insulin; B29-N-(omega-carboxyheptadecanoyl)-des(B30) human insulin and B29-N-(omega-carboxyhepta-decanoyl) human insulin.

(120) Examples of GLP-1, GLP-1 analogues and GLP-1 receptor agonists are, for example, Lixisenatide (Lyxumia®, Exenatide (Exendin-4, Byetta®, Bydureon®, a 39 amino acid peptide which is produced by the salivary glands of the Gila monster), Liraglutide (Victoza®), Semaglutide, Taspoglutide, Albiglutide (Syncria®), Dulaglutide (Trulicity®), rExendin-4, CJC-1134-PC, PB-1023, TTP-054, Langlenatide/HM-11260C, CM-3, GLP-1 Eligen, ORMD-0901, NN-9924, NN-9926, NN-9927, Nodexen, Viador-GLP-1, CVX-096, ZYOG-1, ZYD-1, GSK-2374697, DA-3091, MAR-701, MAR709, ZP-2929, ZP-3022, TT-401, BHM-034. MOD-6030, CAM-2036, DA-15864, ARI-2651, ARI-2255, Exenatide-XTEN and Glucagon-Xten.

(121) An example of an oligonucleotide is, for example: mipomersen sodium (Kynamro®), a cholesterol-reducing antisense therapeutic for the treatment of familial hypercholesterolemia.

(122) Examples of DPP4 inhibitors are Vildagliptin, Sitagliptin, Denagliptin, Saxagliptin, Berberine.

(123) Examples of hormones include hypophysis hormones or hypothalamus hormones or regulatory active peptides and their antagonists, such as Gonadotropine (Follitropin, Lutropin, Choriongonadotropin, Menotropin), Somatropine (Somatropin), Desmopressin, Terlipressin, Gonadorelin, Triptorelin, Leuprorelin, Buserelin, Nafarelin, and Goserelin.

(124) Examples of polysaccharides include a glucosaminoglycane, a hyaluronic acid, a heparin, a low molecular weight heparin or an ultra-low molecular weight heparin or a derivative thereof, or a sulphated polysaccharide, e.g. a poly-sulphated form of the above-mentioned polysaccharides, and/or a pharmaceutically acceptable salt thereof. An example of a pharmaceutically acceptable salt of a poly-sulphated low molecular weight heparin is enoxaparin sodium. An example of a hyaluronic acid derivative is Hylan G-F 20 (Synvisc®), a sodium hyaluronate.

(125) The term “antibody”, as used herein, refers to an immunoglobulin molecule or an antigen-binding portion thereof. Examples of antigen-binding portions of immunoglobulin molecules include F(ab) and F(ab')₂ fragments, which retain the ability to bind antigens. The antibody can be polyclonal, monoclonal, recombinant, chimeric, de-immunized or humanized, fully human, non-human, (e.g., murine), or single chain antibody. In some embodiments, the antibody has effector function and can fix a complement. In some embodiments, the antibody has reduced or no ability to bind an Fc receptor. For example, the antibody can be an isotype or subtype, an antibody fragment or mutant, which does not support binding to an Fc receptor, e.g., it has a mutagenized or deleted Fc receptor binding region. The term antibody also includes an antigen-binding molecule based on tetravalent bispecific tandem immunoglobulins (TBTI) and/or a dual variable region antibody-like binding protein having cross-over binding region orientation (CODV).

(126) The terms “fragment” or “antibody fragment” refer to a polypeptide derived from an antibody polypeptide molecule (e.g., an antibody heavy and/or light chain polypeptide) that does not comprise a full-length antibody polypeptide, but that still comprises at least a portion of a full-length antibody polypeptide that is capable of binding to an antigen. Antibody fragments can comprise a cleaved portion of a full length antibody polypeptide, although the term is not limited to such cleaved fragments. Antibody fragments that are useful include, for example, Fab fragments, F(ab')₂ fragments, scFv (single-chain Fv) fragments, linear antibodies, monospecific or multispecific antibody fragments such as bispecific, trispecific, tetraspecific and multispecific antibodies (e.g., diabodies, triabodies, tetrabodies), monovalent or multivalent antibody fragments such as bivalent, trivalent, tetravalent and multivalent antibodies, minibodies, chelating recombinant antibodies, tribodies or bibodies, intrabodies, nobodies, small modular

immunopharmaceuticals (SMIP), binding-domain immunoglobulin fusion proteins, camelized antibodies, and VHH containing antibodies. Additional examples of antigen-binding antibody fragments are known in the art.

(127) The terms “Complementarity-determining region” or “CDR” refer to short polypeptide sequences within the variable region of both heavy and light chain polypeptides that are primarily responsible for mediating specific antigen recognition. The term “framework region” refers to amino acid sequences within the variable region of both heavy and light chain polypeptides that are not CDR sequences, and are primarily responsible for maintaining correct positioning of the CDR sequences to permit antigen binding. Although the framework regions themselves typically do not directly participate in antigen binding, as is known in the art, certain residues within the framework regions of certain antibodies can directly participate in antigen binding or can affect the ability of one or more amino acids in CDRs to interact with antigen.

(128) Examples of antibodies are anti PCSK-9 mAb (e.g., Alirocumab), anti IL-6 mAb (e.g., Sarilumab), and anti IL-4 mAb (e.g., Dupilumab).

(129) Pharmaceutically acceptable salts of any API described herein are also contemplated for use in a drug or medicament in a drug delivery device. Pharmaceutically acceptable salts are for example acid addition salts and basic salts.

(130) Those of skill in the art will understand that modifications (additions and/or removals) of various components of the APIs, formulations, apparatuses, methods, systems and embodiments described herein may be made without departing from the full scope and spirit of the present invention, which encompass such modifications and any and all equivalents thereof.

Claims

1. A medicament delivery device comprising: a needle configured to move between a retracted position and an inserted position; an insertion mechanism configured to move the needle from the retracted position to the inserted position when activated; a locking arm which, in a locked position, is engaged with the insertion mechanism to prevent activation of the insertion mechanism and, in an unlocked position, is disengaged from the insertion mechanism; a retraction mechanism configured to move the needle from the inserted position to the retracted position when activated; and a trigger configured to move between a disengaged position and an engaged position, wherein the trigger is arranged to be moved from the disengaged position to the engaged position when the medicament delivery device is placed on a body of a patient, and on moving from the disengaged position to the engaged position, the trigger is configured to move the locking arm from the locked position to the unlocked position, and wherein, on moving from the engaged position to the disengaged position, the trigger is configured to activate the retraction mechanism.

2. The medicament delivery device of claim 1, wherein the locking arm is slidably mounted and is pivotably coupled to an unlocking lever such that a rotation of the unlocking lever causes a sliding movement of the locking arm.

3. The medicament delivery device of claim 2, wherein the unlocking lever is pivotably mounted at a midpoint, and comprises a first end pivotably coupled to the locking arm and a second end arranged to engage with the trigger.

4. The medicament delivery device of claim 3, wherein the unlocking lever is formed having an angle at the midpoint, and is arranged such that movement of the trigger along a first direction causes a rotation of the first end, and a resulting rotation of the second end causes a sliding movement of the locking arm along a second direction different to the first direction.

5. The medicament delivery device of claim 1, wherein the trigger, on moving from the disengaged position to the engaged position, is configured to engage with an unlocking lever to rotate the unlocking lever, such that the locking arm is moved from the locked position to the unlocked position.

6. The medicament delivery device of claim 1, wherein: the insertion mechanism comprises a user actuator arranged to be pushed by a user to activate the insertion mechanism; and in the locking position, an end of the locking arm is disposed within a receiving notch formed in the user actuator to prevent movement of the user actuator in a direction perpendicular to the locking arm and, in the unlocked position, the locking arm is withdrawn from the receiving notch.
7. The medicament delivery device of claim 6, wherein the insertion mechanism comprises: a biasing element configured to urge the needle towards the inserted position; and a blocking element arranged, in a blocking position, to prevent the needle from moving from the retracted position to the inserted position and arranged, in a release position, to allow the needle to move from the retracted position to the inserted position, wherein the user actuator is arranged to urge the blocking element from the blocking position to the release position when pushed.
8. The medicament delivery device of claim 7, wherein the insertion mechanism comprises: an injection arm pivotably mounted at one end and pivotably coupled to the needle at the other end; and a guide member configured to restrict movement of the needle to a linear movement between the retracted position and the inserted position in response to a pivoting action of the injection arm, wherein the blocking element in the blocking position is arranged in abutment with the injection arm to prevent movement of the needle to the inserted position.
9. The medicament delivery device of claim 8, wherein: the blocking element is formed having a substantially triangular shape which is pivotably mounted at a first corner to rotate between the blocking position and the release position; and a second corner of the blocking element is in abutment with the injection arm when the blocking element is in the blocking position, and the user actuator is configured to apply a force at a third corner of the blocking element when pushed.
10. The medicament delivery device of claim 8, wherein: the biasing element of the insertion mechanism comprises a coil spring fixed around a pivot of the injection arm; and a first end of the coil spring is fixed in position, and a second end of the coil spring extends along the injection arm to push the injection arm towards the inserted position.
11. The medicament delivery device of claim 8, wherein the retraction mechanism comprises a return actuator pivotably mounted to rotate around the pivot of the injection arm between a first position and a second position, and the return actuator in the first position is arranged to abut with the injection arm in the inserted position, and the return actuator is configured to urge the injection arm and the needle to the retracted position of the needle when moved to the second position.
12. The medicament delivery device of claim 11, wherein the blocking element, in the blocking position, is engaged with the return actuator to retain the return actuator in the first position and, in the release position, is disengaged from the return actuator.
13. The medicament delivery device of claim 12, wherein: the trigger is formed having a retaining slot; and in the first position, the return actuator is aligned with the retaining slot, disposed within the retaining slot, and prevented from moving to the second position when the trigger is in the engaged position, and is released by the retaining slot when the trigger moves from the engaged position to the disengaged position; and in the second position, the return actuator is not aligned with the retaining slot and abuts with the trigger to prevent the trigger from moving from the disengaged position to the engaged position.
14. The medicament delivery device of claim 11, wherein the retraction mechanism comprises a biasing element configured to urge the return actuator towards the second position, wherein the trigger in the engaged position is configured to engage with the return actuator to retain the return actuator in the first position, and the trigger in the disengaged position is disengaged from the return actuator.
15. The medicament delivery device of claim 14, wherein the biasing element of the retraction mechanism comprises a coil spring, having a first end of the coil spring fixed in position and a second end of the coil spring in abutment with the return actuator to push the return actuator towards the second position, and wherein the coil spring is not fixed in position between the first

end and second end.

16. The medicament delivery device of claim 15, wherein the trigger is pivotably mounted to rotate between the engaged position and the disengaged position, and wherein the first end of the coil spring of the retraction mechanism is fixed to the trigger to bias the trigger towards the disengaged position.

17. The medicament delivery device of claim 1, comprising an outer housing, wherein the needle is disposed within the outer housing in the retracted position and extends out of the outer housing through a contact surface of the outer housing in the inserted position, wherein the trigger extends out of the outer housing through the contact surface of the outer housing in the disengaged position, wherein the trigger in the disengaged position extends beyond the needle in the inserted position, and wherein the trigger is configured to be moved from the disengaged position to the engaged position when the contact surface is placed against a user.

18. The medicament delivery device of claim 17, wherein a length of the outer housing, measured perpendicular to the contact surface, is smaller than a width of the outer housing.

19. A medicament delivery device comprising: a housing; a needle configured to move relative to the housing between a retracted position and an inserted position; a locking arm configured to move relative to the housing from a first position in which movement of the needle from the retracted position to the inserted position is prevented to a second position in which movement of the needle from the retracted position to the inserted position is allowed; and a trigger element configured to move relative to the housing from an extended position to a retracted position, the medicament delivery device configured such that (i) movement of the trigger element from the extended position to the retracted position causes the locking arm to move from the first position to the second position and (ii) movement of the trigger element from the retracted position to the extended position causes the needle to move from the inserted position to the retracted position.

20. A method comprising: moving a trigger of a medicament delivery device relative to a housing of the medicament delivery device from a first position in which a locking arm is engaged with an insertion mechanism of the medicament delivery device to prevent movement of a needle relative to the housing from a retracted position to an inserted position to a second position in which the locking arm is disengaged from the insertion mechanism, wherein when the trigger is in the second position, the insertion mechanism moves the needle relative to the housing from the retracted position to the inserted position, and after moving the needle relative to the housing from the retracted position to the inserted position, the trigger moves relative to the housing from the second position to the first position causing the needle to be moved by a retraction mechanism of the medicament delivery device from the inserted position to the retracted position.
